

Research Paper

Beyond sentiment: Using large language models to decode multidimensional urban park perceptions for enhanced equality

Qiwei Song^{a,b,c}, Siyu Tian^{c,d}, Lingwei Zheng^{c,d}, Yuxuan Zheng^{c,e},
Lin Qiu^{f,e,g}, Bo Huang^h, Jeroen van Ameijde^{a,*}

^a School of Architecture, The Chinese University of Hong Kong, Hong Kong SAR, China

^b Senseable City Laboratory, Department of Urban Studies and Planning, Massachusetts Institute of Technology, Cambridge, MA, USA

^c Computational Social Science Laboratory, Faculty of Social Science, The Chinese University of Hong Kong, Hong Kong SAR, China

^d Department of Geography and Resource Management, The Chinese University of Hong Kong, Hong Kong SAR, China

^e Department of Psychology, The Chinese University of Hong Kong, Hong Kong SAR, China

^f School of Journalism and Communication, The Chinese University of Hong Kong, Hong Kong SAR, China

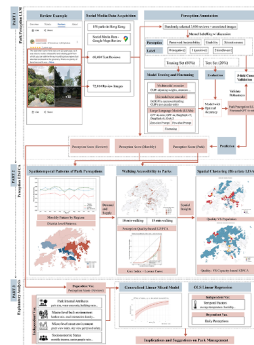
^g School of Governance and Policy Science, The Chinese University of Hong Kong, Hong Kong SAR, China

^h Department of Geography, The University of Hong Kong, Hong Kong SAR, China

HIGHLIGHTS

- We systematically benchmarked various deep learning algorithms to interpret perceived park qualities from online reviews.
- Our fine-tuned Park-Perception-LLM achieved an accuracy from 83 % to 91 %.
- Perception metrics were integrated into an E2SFCA framework to model park accessibility from a quality perspective.
- Inequalities in park quality versus quantity, and spatial mismatch patterns are revealed.
- Spatiotemporal factors that inform park perceptions are uncovered using multi-variate regression.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Landscape perception
Walking accessibility
Natural Language Processing (NLP)
Large Language Model (LLM)
Social equality
Environmental justice
Social media data
Machine learning

ABSTRACT

Understanding how people value urban parks is vital for designing liveable and equitable cities, yet the nuanced measurement of park perception proves elusive. While analysis of user-generated content from social media offers scalability over conventional small-sample surveys or objective indicators, deriving effective perceptions beyond basic sentiment, aligned with actionable urban planning variables, remains a crucial challenge. Consequently, a critical gap persists in evaluating inequality in park accessibility through the lens of multi-dimensional quality experiences within diverse urban settings. This hinders truly citizen-centric green space planning and the derivation of insights that transcend quantity. To address this challenge, this study pioneers a systematic benchmarking of large language models (LLMs) and other encoders to classify three perception dimensions—perceived accessibility, usability, and attractiveness—from massive online review data. The fine-tuned Park-Perception-LLM achieved 83–91% accuracy, outperforming other algorithms. Applied to Hong Kong as a representative case of a high-density metropolis, we measured perception scores for 158 urban parks and

* Corresponding author.

developed a novel perception-weighted park accessibility metric. This uncovered pronounced inequalities, identifying neighbourhoods with abundant park capacity yet poor experiential quality—a nuanced socio-spatial disparity. Subsequent analysis of perception pathways revealed that diverse facility provision, natural park elements, nearby cultural amenities, and green streetscapes could significantly enhance perceived qualities, whereas monotonous urban forms undermine attractiveness. This LLM-driven framework accurately infers comprehensive perceptions from online reviews, providing a scalable analytical foundation to support planners, designers, and policymakers in integrating perception-based evidence into future park planning, design, and management decisions towards better quality of life.

1. Introduction

Urban parks are critical green infrastructure for sustainable urbanism, improving urban ecological systems and climate characteristics, and offering vital resources for recreation (Boutreux et al., 2024). Extensive empirical evidence has suggested that parks confer multifaceted benefits, including enhancing mental well-being (Callaghan et al., 2021), stimulating activity to improve physical health (Richardson et al., 2013), and facilitating social mixing to mitigate segregation (Heine et al., 2025). As over two-thirds of the global population is expected to reside in cities by 2050, ensuring equitable access to high-quality green spaces for all has become an urgent priority. This echoes the United Nations Sustainable Development Goal 11.7, to provide safe and inclusive public places for all generations (United Nations, 2015), leading to an increased need for evidence-based, data-driven assessments of green space equality (Battiston & Schifanella, 2024).

In contrast to planning guidelines setting static targets such as the ‘3-30-300’ rule (Konijnendijk, 2023), urban scholars now employ fine-grained geospatial models that capture distance to and capacity of parks (Y. Fu et al., 2024; Liang et al., 2023) to measure the ‘accessibility’ concept (Geurs & van Wee, 2004). Nevertheless, such metrics remain narrowly quantity-based, overlooking multi-dimensional qualities that drive visitation (Battiston & Schifanella, 2024; Wang et al., 2022). Where quality is considered at all, it is often simplified to coarse proxies such as park acreage (Cheng et al., 2019; Zou et al., 2024). Recent work has begun to operationalise more nuanced measures objectively, including grass cover, water features, and amenity provision, analysed through remote-sensing and spatial data (Biernacka et al., 2022; Guo et al., 2024; Rigolon, 2016; Stessens et al., 2020). Nonetheless, such an exclusive focus on “objective” quality is problematic. Mounting evidence suggests that objectively measured park conditions often diverge from users’ subjective lived experiences, casting doubt on whether researchers’ quality metrics are an accurate representation of public expectations (Liu et al., 2024). Capturing subjective perceptions is therefore instrumental if policymakers aim to align green space provision with day-to-day preferences and needs to advance urban liveability (Biernacka et al., 2022; Sheikh & van Ameijde, 2022).

1.1. Review of perception studies

A wealth of environmental psychology and landscape perception research has established that human experience of landscapes is multi-dimensional, encompassing cognitive, affective, and behavioural processes. For instance, the widely recognised Attention Restoration Theory by Kaplan & Kaplan (1989) posits that exposure to nature restores cognitive resources through four perceptual dimensions—being away, extent, fascination, and compatibility—which explain how natural settings alleviate mental fatigue by fostering effortless attention and immersive engagement. Likewise, Appleton (1975)’s Prospect–Refuge Theory emphasises the evolutionary basis of landscape aesthetic preference, proposing that people are drawn to environments offering prospect (open views and visibility) and refuge (enclosure and safety). Later interpretations expanded to include perceptions of mystery and visual complexity (Dosen & Ostwald, 2016), aligning with the Complexity–Arousal Model, which proposes that aesthetic preference follows

an inverted-U relationship with visual complexity to achieve arousal (stimulation evoked by the scene) (Berlyne, 1971).

While these classical theories primarily concern natural landscapes, urban planning and design research has extended perception studies to the built environment, focusing on street-level experiences. Particularly, through the employment of a panel of experts, Ewing & Handy (2009) identified several key perceptual dimensions of walkable environments, including human scale, enclosure, transparency, complexity, and imageability. Similarly, adopting an image-based survey approach, scholars employed crowd-sourcing platforms to develop the MIT Place Pulse 2.0 dataset, which enabled large-scale assessments of six subjective perceptions—safe, beautiful, boring, wealthy, depressing, and lively—based on global street views (Dubey et al., 2016). Collectively, these classical and contemporary frameworks underscore that perception arises from the dynamic interplay between environmental structure and psychological response, highlighting the importance of understanding how people perceive the urban realm to reflect their lived experiences.

1.2. Progress in quantifying park perceptions

Nevertheless, measuring subjective perceptions of urban parks at the metropolitan scale is challenging. Conventional methods rely on structured questionnaires and in-person interviews, which yield rich insights but are labour-intensive, costly, and time-consuming, rendering city-wide monitoring impractical (Moran et al., 2020; Stessens et al., 2020). The rise of crowd-sourced social media data (SMD) offers a promising alternative. Platforms such as Google Maps, Instagram, and X (formerly Twitter) transform citizens into social sensors, offering real-time, geo-tagged opinions at marginal cost (Liu et al., 2015). Among the three primary forms of human-centred big data widely adopted in related literature are street view imagery (SVI), mobility traces, and SMD. It is found that the text-rich SMD is uniquely suitable for uncovering human interactions with parks and subjective preferences for open space (Li et al., 2024). Nevertheless, it is worth noting that SMD is not drawn from a random sample and inherently suffers from representativity biases, as user participation is shaped by self-selection and unequal access to digital platforms (Hargittai, 2020). Such user-generated content typically reflects the behaviours and preferences of active online users with stronger attitudinal tendencies than the general population, underrepresenting groups such as older adults and children who have limited internet access (Heikinheimo et al., 2020).

Technically, two major natural language processing (NLP) pipelines dominate current practice in analysing unstructured SMD text contents. First, lexicon-based methods match words to a sentiment dictionary and tally their frequency to infer emotional tendency (Kong et al., 2022; Tang et al., 2023). Second, deep learning models embed entire sentences and apply transformer architectures such as Bidirectional Encoder Representations from Transformers (BERT), outperforming earlier Long Short-Term Memory (LSTM) and other conventional models in processing large corpora (Huai et al., 2023; Zhao et al., 2024). More recently, advanced generative large language models (LLMs) like Generative Pre-training Transformer (GPT) adopt an autoregressive (decoder-only or encoder-decoder) architecture that explicitly models the probability of the next token (Brown et al., 2020). With vastly

massive training corpora and reduced computational costs, scaling laws endow LLMs with an exceptional ability to generalise across domains and to infer nuanced theoretical constructs from unstructured text, a capacity already demonstrated in social science research (Ziems et al., 2024). It suggests considerable untapped potential for urban studies applications, including interpreting subjective perceptions to measure park quality and inform evidence-based planning at scale.

1.3. Research gaps

Despite these advancements in using SMD and LLMs for text analysis, several critical knowledge gaps persist. First, although SMD has become a valuable lens for capturing users' subjective opinions, existing studies have primarily focused on emotions or sentiment analysis (Huai & Van de Voorde, 2022; Kong et al., 2022; Tang et al., 2023; Zhou et al., 2024). A smaller subset of research has incorporated computer vision to analyse user-generated images—such as Flickr photos—to identify park-related themes (Ghermandi et al., 2022; Li et al., 2024). However, while these approaches have enriched our knowledge in important ways, the narrow focus on emotional expressions or the classification of visual content fails to establish systematic connections between perceptions and environmental characteristics, thereby limiting the potential to derive actionable insights for design and management improvements (Zhao et al., 2025). Nuanced perceptions that directly indicate spatial or management qualities, including perceived accessibility, usability, and attractiveness, remain largely unmeasured, leaving gaps in understanding the perceived value of park use (Biernacka et al., 2022; Moran et al., 2020). Given the advancements in LLMs and their yet nascent use in planning research (X. Fu et al., 2024), no study has yet benchmarked their ability to infer nuanced park perceptions from SMD. Consequently, there is an urgent need for a generalisable method to support further analysis of multi-dimensional park perceptions.

Moreover, far too little attention has been paid to revealing equality in perceived park qualities. Current work typically maps perceived quality but overlooks whether residents can actually reach those parks (Guo et al., 2024; Huai et al., 2023). In contrast, the Enhanced Two-Step Floating Catchment Area (E2SFCA) models can jointly account for park supply and population demand across various travel modes (Tao et al., 2023) and are now commonly used in capacity-based greenspace accessibility studies (Cheng et al., 2019; Xing et al., 2020). Nevertheless, very few studies integrate revealed park perceptions; those that do simplify focus on sentiment alone (Zou et al., 2024) or on objective park attributes (e.g., water ratio) (Y. Fu et al., 2024; Xing et al., 2020) or park size (Adorno et al., 2025; Liang et al., 2023) using an E2SFCA metric to reflect the accessibility to quality parks. Policies derived purely from capacity-based accessibility measures risk overlooking experiential disparities to reveal residents' unmet needs. Integrating perception-based measurements into the existing E2SFCA accessibility framework could help expose mismatches between planned quantity and experienced quality (lived experience), relationships that are rarely covered in previous studies due to data constraints.

Finally, how the spectrum of park perceptions is shaped by the macro-scale built form, micro-scale streetscapes, and temporal attributes has not been well documented. Existing work primarily links environmental variables to sentiment (Huai & Van de Voorde, 2022; Kong et al., 2023; Tang et al., 2023) or usage (Moran et al., 2020), while neglecting other perceptual dimensions. Although topic modelling and text frequency analysis provide preliminary thematic clues (Kong et al., 2023; Wan et al., 2021), rigorous statistical modelling is needed to reveal the mechanisms that underpin perceptions for targeted interventions (Huai & Van de Voorde, 2022; Tang et al., 2023). Moreover, temporal influences such as seasonality and temperature have received little attention (Zhao et al., 2025), leaving dynamic patterns in park perceptions largely uncharted. All in all, these identified shortcomings underscore the need for more sophisticated, multi-dimensional approaches to understanding and planning urban park systems.

1.4. Research Aims

This study addresses these gaps through three key questions:

(1) In what ways can domain-specific knowledge be leveraged to enhance the fine-tuning of large language models for the comprehensive assessment of park perception dimensions?

(2) How can we evaluate people's access to high-quality parks to uncover inequalities and potential spatial mismatches between park capacity and quality?

(3) What are the spatiotemporal factors that shape individuals' multifaceted perceptions of parks?

The research questions are addressed through a structured analytical workflow (Fig. 1). First, various advanced deep learning model workflows (i.e., multimodal encoders, text-only encoders, and LLMs) were developed and tested for their capacity to predict three park quality dimensions (perceived accessibility, usability, and attractiveness) from a manually labelled set of Hong Kong-based Google Maps reviews (Sections 3.1 and 4.1). The best-performing model was then applied to over 68,400 reviews, generating multi-dimensional perceptions for 158 urban parks.

Second, these perception scores were processed through a proposed perception-weighted walking accessibility framework that evaluates multiple distance thresholds, assessing spatial equality to pinpoint neighbourhoods where experienced quality diverges from supplied capacity, thereby improving the interpretation of park accessibility measures. The accessibility modelling and spatial mismatch analysis are presented in Sections 3.2 and 4.2 - 4.5, which integrate perceptions of park quality with population catchment and service coverage measures.

Third, multivariate regressions were conducted to reveal the effects of different characteristics, including park interior features (e.g., water ratio), macro-level built environment attributes (e.g., amenities), micro-level street qualities (e.g., greenery), as well as the nearby sociodemographic profiles and temporal factors on various perception dimensions, given that their influencing mechanism has been so far poorly understood. The spatiotemporal regression and heterogeneity analyses are detailed in Sections 3.3, 4.6, and 4.7, exploring how individual perceptions vary across time, location, and built-environment contexts.

Together, these analyses provide an integrated understanding of both the methodological advances in perception modelling and their implications for spatial equity and urban park planning.

2. Study area

Hong Kong is a compact, high-density metropolis, with over 7.5 million residents (as of 2023) concentrated in about 25% of its 1,110 km² territory, with the remaining protected as conservation land (C&SD., 2024). The city comprises Hong Kong Island, Kowloon, and the New Territories, which are further divided into 18 administrative districts (Fig. 2). Due to its unique socio-political context, spatial planning remains a largely top-down process, where new open spaces are disproportionately sited in premium commercial or mixed-use precincts rather than ordinary residential neighbourhoods, deepening existing inequities in park provision (Tang, 2017). Hong Kong's Planning Standards and Guidelines still emphasise minimum per-capita areas of recreational space and pay little attention to qualitative dimensions such as maintenance, convenience, aesthetics, and comfort (Sheikh & van Ameijde, 2022). However, such experiential qualities significantly impact people's behavioural responses to the built environment, for which higher perceived accessibility translates into longer and more frequent park visits (Lau et al., 2021). Park quality rather than proximity has been shown to predict use statistics in the context of high-density urban planning (Wang et al., 2022). These shortcomings underscore the urgent need to transcend quantity-based planning and examine whether residents can equitably access high-quality parks by considering multidimensional perceptions.

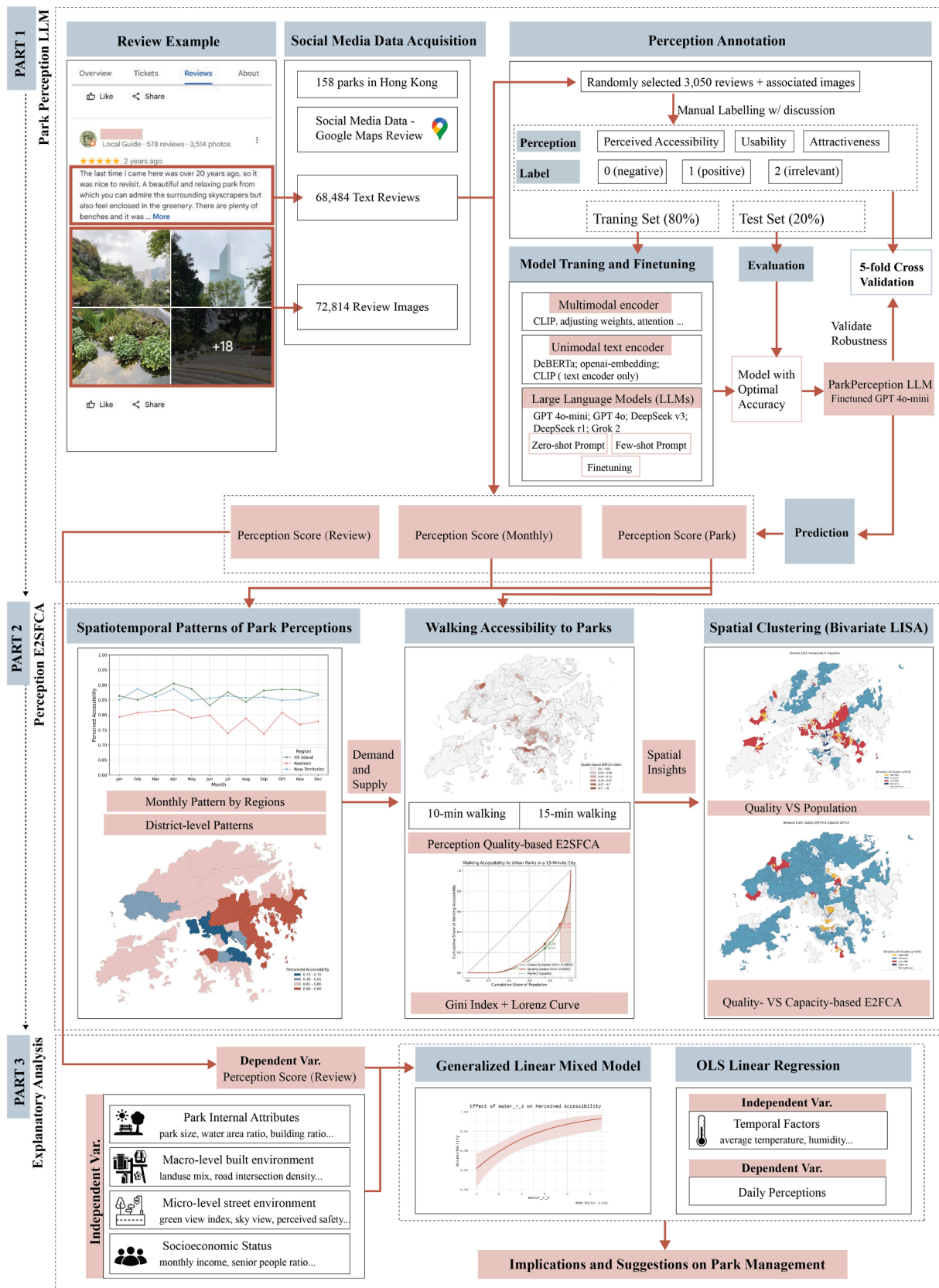


Fig. 1. Analytical framework.

The New Territories

- 1 Islands
- 2 Kwai Tsing
- 3 North
- 4 Sai Kung
- 5 Sha Tin
- 6 Tai Po
- 7 Tuen Wan
- 8 Tuen Mun
- 9 Yuen Long

Kowloon

- 10 Kowloon City
- 11 Kwun Tong
- 12 Sham Shui Po
- 13 Wong Tai Sin
- 14 Yau Tsim Mong

Hong Kong Island

- 15 Central & Western
- 16 Eastern
- 17 Southern
- 18 Wan Chai

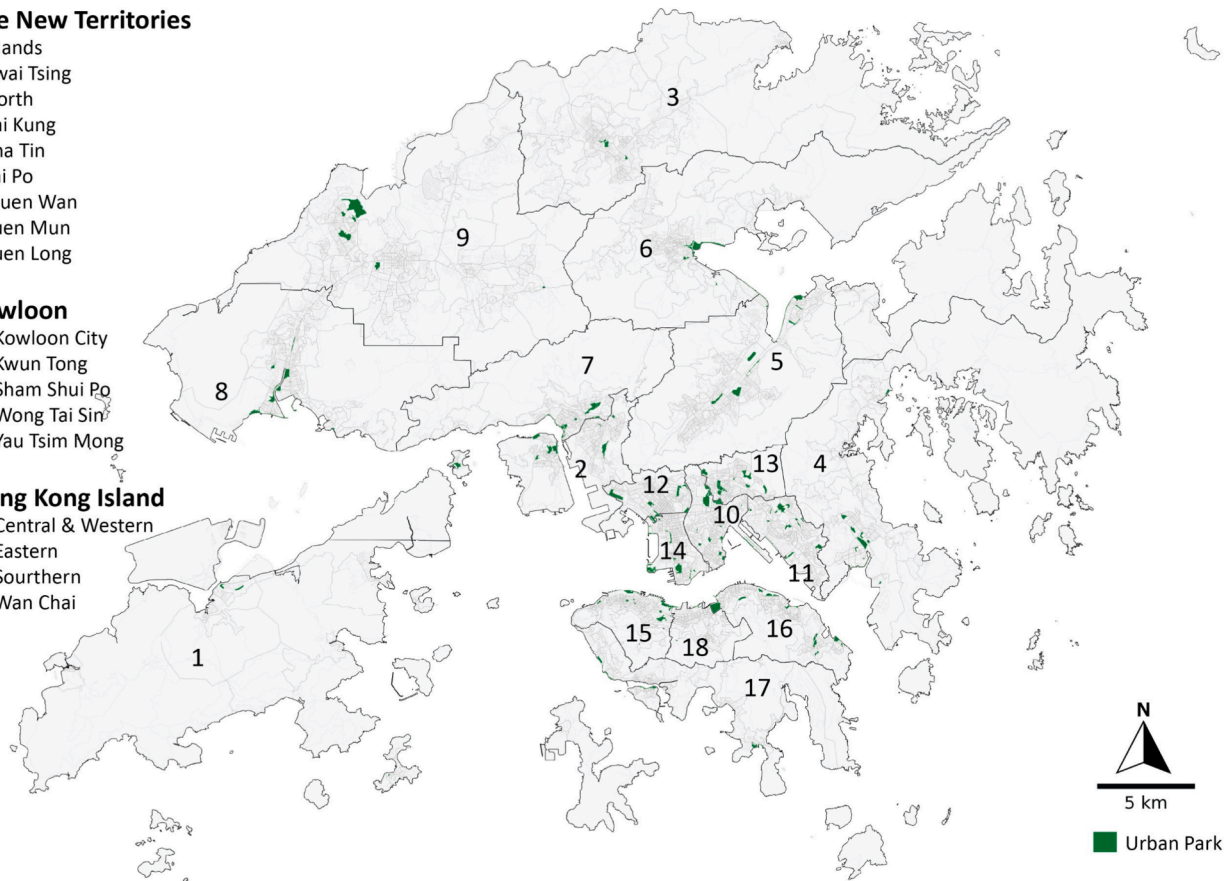


Fig. 2. Hong Kong's administrative districts and urban park locations.

2.1. Park perceptions of interest

Understanding park perceptions requires a multidimensional framework that captures how individuals experience and evaluate urban green spaces beyond what objective measures can reveal. Drawing upon both international and regional green space literature (Biernacka & Kronenberg, 2018; Zhang & Tan, 2019; Zhu et al., 2025), this study focuses on three interrelated and complementary dimensions—perceived accessibility, usability, and attractiveness—to comprehensively represent users' subjective evaluations of urban parks.

The selection of these three dimensions is theoretically grounded in the urban green space provision framework proposed by Biernacka & Kronenberg (2018), which conceptualises park provision in three distinct levels: *availability*, *accessibility*, and *attractiveness*.

The notion of *availability* refers to the existence or spatial presence of green spaces before considering other experiential dimensions (Biernacka & Kronenberg, 2018). In practice, this aspect is already well captured by proximity-based accessibility metrics, as distance-decay and capacity-based models inherently account for the spatial availability of parks and the supply–demand relationship between parks and users (Adorno et al., 2025). As this notion represents a precondition of access, our framework meaningfully further extends it by integrating other dimensions to eventually form a perception-informed accessibility measure.

The notion of *accessibility* extends beyond physical distance to include subjective ease of access. This perceived accessibility refers to a visitor's subjective judgement of the ease of reaching a park, shaped by distance, travel time, transport options, and perceived barriers (Biernacka et al., 2022; Moran et al., 2020). Empirical studies further underscore the significance of this distinction: in Singapore and Australia, perceived accessibility explained park visitation better than

objective distance measures (Wang et al., 2015; Zhang & Tan, 2019), and nationwide surveys also identified it as the second-strongest predictor of park use after people's connectedness to nature (Sia et al., 2023).

Biernacka & Kronenberg (2018)'s notion of *attractiveness* is operationalised for this study by referring to recent park quality studies. As Rigolon (2016) has critically argued, park equity entails not only proximity but also the quality of facilities, maintenance, and safety perceptions. De la Barrera et al. (2016) conceptualised the “image” perception of green spaces as a combination of perceived functionality and the symbolic meaning they fulfil. Zhao et al. (2024) further examined the perceptions of usability and attractiveness in parks using SMD, thereby complementing existing accessibility-focused analyses and capturing subjective user opinions from experiential perspectives. Similarly, the present study introduces two additional perception dimensions for operationalisation: usability and attractiveness, which can effectively reflect these qualitative aspects. Usability captures how well a park fulfils functional expectations, reflected in the availability, adequacy, and condition of its provided facilities (Moran et al., 2020; Zhao et al., 2024). Attractiveness denotes a park's overall appeal—size, aesthetic quality, natural elements, cultural features, and supporting services—which motivates voluntary visits and extended stays as suggested by the ‘push-pull’ theory (Biernacka et al., 2022; Zhan et al., 2024). A similar composite quality index was constructed by Zhang et al. (2022) based on aesthetic features and recreational and convenience facilities, utilising social media imagery, which demonstrates that the dimensions of usability and attractiveness serve as effective indicators of overall park quality.

3. Data & methods

3.1. Fine-tuning models for perception prediction

3.1.1. SMD acquisition

Compared to visitor-oriented platforms such as TripAdvisor and social platforms such as Twitter (Huai et al., 2023; Kong et al., 2022), Google Maps reviews are more tightly linked to specific places and amenities, thereby offering insights into locals' place-based experiences and sentiments (Huai & Van de Voorde, 2022; Zhu et al., 2025). According to a 2021 industry report, around 62% of user-generated reviews on Google Maps were contributed by 'Local Guides' — users who voluntarily provide feedback without monetary incentives, instead earning points or badges through Google's Local Guides program (Rollison, 2021). Moreover, the use of these online review data for non-commercial research purposes complies with Google's fair use policy (https://about.google/intl/en-GB_ALL/brand-resource-center/products-and-services/geo-guidelines/).

Based on the Hong Kong Lands Department's iGeoCom 4.0 database, we extracted public amenities type classified as 'PAR' (Park) under the class 'RSF' (Leisure Facility), removed duplicated records, and merged redundant listings. After querying the Google Places Application Programming Interface (API), it yielded 158 officially recognised urban parks (85% larger than 1 ha), each identified by a unique place_id. We retrieved user reviews and images posted over the past ten years (January 2015 – December 2024) to ensure comprehensive data coverage and mitigate potential temporal fluctuations in user activities. It should be noted that 14 parks (8.8% of the total sample) were established after 2015 and contain 1,768 associated reviews (2.5% of the total reviews). Given this relatively small proportion and the use of aggregated multi-year perception measures, the overall perception patterns captured in this study are unlikely to be substantially biased by these late additions. After eliminating invalid records, the final dataset comprised 68,484 text reviews and 72,814 images. A detailed account of the data cleansing workflow is documented in Supplementary Material Appendix A.

3.1.2. Labelling park perceptions

Because no open-source labels exist for perceptions in Hong Kong, we customised our dataset following the established protocol (Zhao et al., 2024). We randomly selected 3,050 comments from the dataset and manually annotated the three perceptions using a three-way code (0 = negative, 1 = positive, 2 = irrelevant) for each comment. To ensure the labels accurately captured local residents' multi-dimensional perceptions, four independent experts were recruited—each possessing an urban planning or design background and having lived in Hong Kong for an extended period. Inspired by prior perception studies (Q. Song et al., 2025), all coders were both familiar with local park experiences and well-versed in professional terminology to minimise potential biases. Prior to labelling, the definitions of the three perception dimensions and sample annotations were explained in detail to standardise their understanding.

To establish a consistent coding standard, all coders first annotated an overlapping subset of 1000 reviews. Any discrepancies in interpretation were discussed collectively, and consensus was reached through detailed deliberation (Zhao et al., 2024). A comprehensive codebook was developed in this process to formalise guidelines for labelling the three perceptions. Furthermore, balancing workload and resources, the coders proceeded to label the remaining comments independently. In case coders encountered labels that they were uncertain about, these labels were revisited and resolved through group discussion again to ensure consistency to the largest extent. This multi-stage labelling procedure ensured a rigorous, reliable, and contextually grounded labelling outcome, ensuring model validity.

3.1.3. Model fine-tuning and selection

To comprehensively benchmark the state-of-the-art algorithms on park perceptions, we compared three model families: (i) multimodal encoders, (ii) unimodal text-only encoders, and (iii) LLMs, adopting an 80%-20% train-test split on the labelled 3050 reviews.

(i) multimodal encoders.

Regarding multimodal encoder models, we fine-tuned a pretrained Contrastive Language-Image Pre-training (CLIP) model with different settings and parameters into four separate models. We adopted 'openai/clip-vit-base-patch16', an open-access multimodal model proposed by OpenAI, pretrained on 400 M image-text pairs to align visual and linguistic embeddings, and fine-tuned it using our customised training data for perception classification (Radford et al., 2021). It effectively learned to match the correct image-text pairs by maximising their similarity and minimising it for mismatched pairs. The fine-tuning process further allowed the fusion of both images and texts for the downstream prediction tasks, which have been used for predicting urban function, perceptual qualities, and depression rates in urban studies (Huang et al., 2024; Ouyang et al., 2024; Zhao et al., 2024), implying its great potential for our tasks using multimodal architecture, which interprets both text and image data. The technical details on parameters are provided in Appendix B.

(ii) unimodal text-only encoders.

The second stream consists of three text-encoder models. We experimented with CLIP-text, where the CLIP architecture muted its image channel to solely rely on text in its prediction. Furthermore, we fine-tuned a Decoding-enhanced BERT with Disentangled Attention (DeBERTa) model, which was reported to outperform BERT in previous classification tasks (He et al., 2021). This BERT-based encoder model has been proven effective in urban studies and has been widely applied to measure sentiment scores, which are also seen as a more advanced benchmark model (Rui, 2023; Zhou et al., 2024) compared to more conventional ML algorithms such as LSTM. Additionally, we tested the recent 'text-embedding-3-large' model by OpenAI, which creates embeddings with up to 3,072 dimensions (OpenAI, 2024), capturing richer semantics that may more effectively predict perception labels.

(iii) LLMs.

Finally, the LLM experiments comprised three commercial versions—GPT-4o-mini, GPT-4o (OpenAI), and Grok-2 (XAI), as well as two open-source alternatives—DeepSeek-v3 and DeepSeek-r1. Compared to BERT-based models, LLMs are significantly larger (billions of parameters) and trained on vast, diverse internet-scale datasets, allowing them to generalise to more open-ended tasks (Raiaan et al., 2024). Such models typically employ decoder-only or encoder-decoder architectures that generate text sequentially, and this human-machine conversational ability has completely revolutionised the NLP field (de Kok, 2025). Although rarely used for park perception classification, LLMs have shown promise in sentiment analysis (Zou et al., 2024) and cultural system service classification (H. Luo et al., 2025).

The models underwent three evaluation tasks through prompt engineering (see Fig. 3 for schematic diagram): (i) Zero-shot learning: direct classification of reviews prompted only with definitions of perceptions (detailed prompt described in Appendix C); (ii) Few-shot learning: To enhance task adaptability, the prompt was further augmented with 20 labelled representative examples extracted from the training set. The few-shot learning technique is a cost-effective approach that transforms large language models from static text generators into dynamic learners, enabling them to adapt to customised tasks using only a few valid examples without retraining (Parnami & Lee, 2022). (iii) Supervised Fine-tuning: The best-performing few-shot model, GPT-4o-mini, was subsequently fine-tuned on the entire training corpus to develop our customised model, Park-Perception-LLM. The fine-tuning was conducted over five epochs with a batch size of 8, a learning-rate multiplier of 1.8, and a random seed of 42. Specifically, the GPT-4o-mini in this paper refers to the pretrained model gpt-4o-mini-2024-07-18, which is a snapshot version of the GPT-4o-mini model

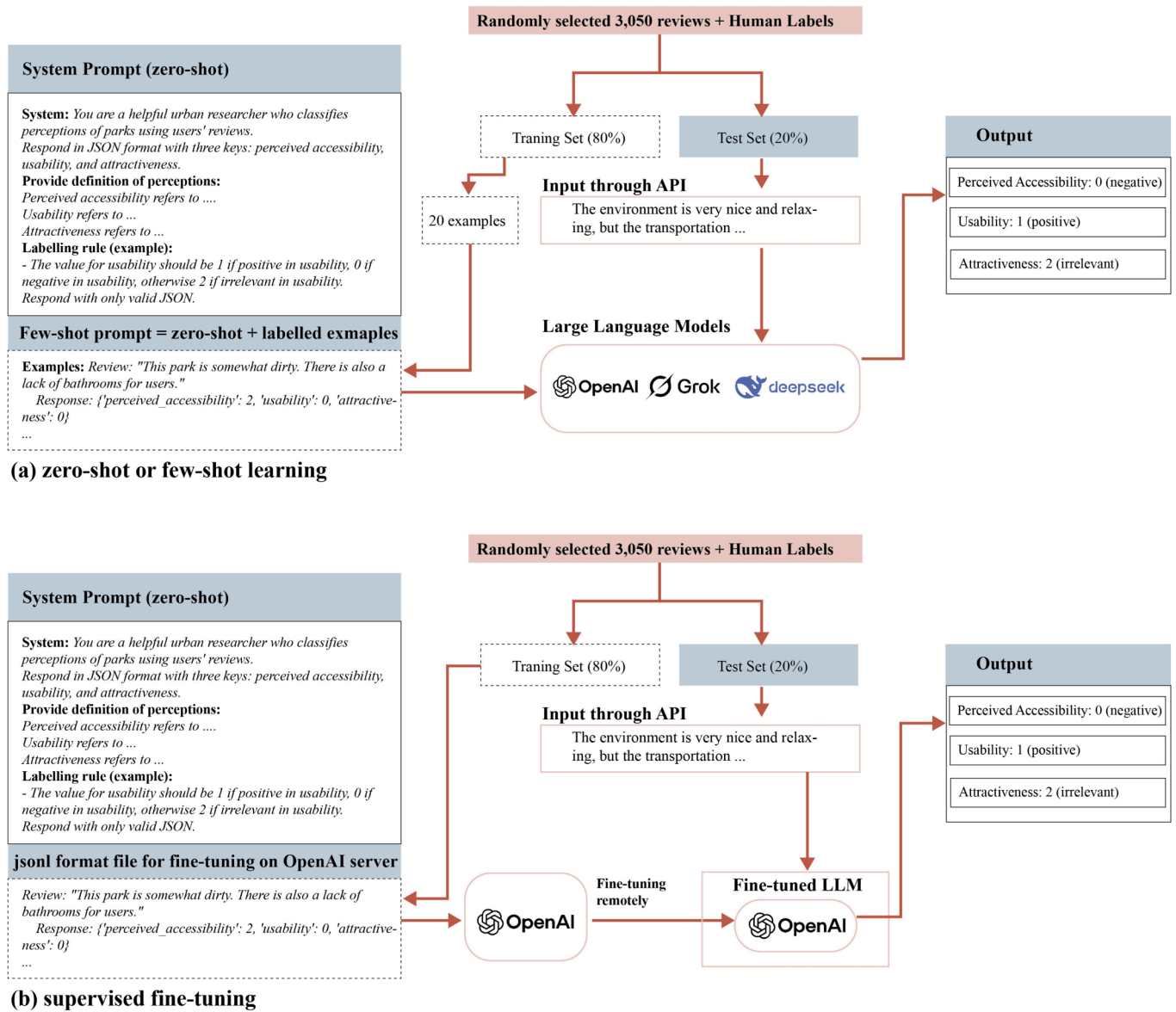


Fig. 3. Schematic diagram of prompt for LLMs.

released on July 18, 2024.

Notably, unlike the fine-tuning processes of the CLIP and DeBERTa models, which were conducted locally, the prompt engineering and fine-tuning of the LLM were implemented via the OpenAI API. Fine-tuning process enhances model consistency, response formatting, and domain adaptation. Given the large volume of park review data, this approach enabled the model to leverage a substantially larger number of examples than would be possible with zero-shot or few-shot prompting alone. Fine-tuning was executed securely via the OpenAI API, where training was performed on remote servers using our uploaded training corpus; model weights were updated only under OpenAI's managed fine-tuning protocol. To the best of our knowledge, this study represents one of the earliest applications of supervised fine-tuning of large language models within urban studies, specifically for advancing data-driven perception research.

All sixteen models were evaluated using the top-1 accuracy (Appendix D) on the hold-out test dataset. To further substantiate model robustness and ensure generalisation across data partitions, a five-fold cross-validation procedure was subsequently applied to the Park-Perception-LLM. This process provides a more rigorous assessment of

predictive stability and mitigates potential bias arising from any single train-test split.

3.2. Revealing spatiotemporal park perceptions

3.2.1. Spatiotemporal patterns

Using the accuracy score (top-1) to compare all candidate LLMs, we identified the best-performing model—Park-Perception-LLM, applied it to the full set of reviews, and aggregated the resulting perceptions to the park level. For each dimension, its perception score equals the share of positive reviews among all relevant reviews, yielding a score of 0–1. To ensure data quality, parks with fewer than 10 comments (~ 5% of parks) were imputed using the mean park perception in the same administrative district for spatial equality analysis. Temporal dynamics were examined through separate monthly and yearly aggregations of reviews to identify seasonal and long-term trends.

3.2.2. Developing perception-based park accessibility

To assess whether parks meet neighbourhood-level needs and integrate park quality with accessibility, we modified the ESFCA

method—widely applied in park equality research (Cheng et al., 2019; Tao et al., 2023)—by replacing the conventional capacity (cumulative park size) with perception-based quality. Because park features are better described in polygon surfaces rather than points, we followed established practice to simulate their entrance locations by intersecting simulated routes with buffered boundaries (Zhang et al., 2023). Therefore, the walking-to-park accessibility metric could be divided into two steps:

Step 1: For each entrance as supply, each centroid of neighbourhood as a demand point within walking distance threshold d_0 , and the supply-demand ratio R_j at the entrance j is:

$$R_j = \frac{S_j}{\sum_{k \in \{d_k \leq d_0\}} P_k G_{k,j}} \quad (1)$$

where S_j reflects the supply at the entrance j , which in prior studies typically use park size, in this research can refer to formula (3); P_k denotes the population in community k that can reach park entrance j using distance d_k , which is below the d_0 ; $G_{k,j}$ demonstrates the distance impedance coefficient to calculate the weights.

$$G_{k,j} = \begin{cases} \frac{e^{-\left(\frac{1}{2}\right) \times \left(\frac{d_k}{d_0}\right)} - e^{-\left(\frac{1}{2}\right)}}{1 - e^{-\left(\frac{1}{2}\right)}}, & d_k \leq d_0 \\ 0, & d_k > d_0 \end{cases} \quad (2)$$

$$S_j = S_{access} + S_{usabil} + S_{attract} \quad (3)$$

where S_{access} denotes the perceived accessibility; S_{usabil} represents the usability; $S_{attract}$ refers to the attractiveness, reflecting the aggregated park qualities on the supply side (Zhao et al., 2025).

Step 2: For community i , the calculated walking accessibility to park entrance A_i can sum up the supply-demand ratios R_p weighted by the distance decay coefficient $G_{i,p}$:

$$A_i = \sum_{p \in \{d_{i,p} \leq d_0\}} G_{i,p} R_p \quad (4)$$

In the perception-weighted park accessibility, each community is linked to only the closest entrance of every park, mirroring residents' tendency to choose the quickest entry and avoiding overcounting. Guided by empirical evidence, we adopted 800 m (~ 10 min) and 1,200 m (~ 15 min) network buffers (Tao et al., 2023). Origin-destination distances using the shortest network path were measured in ArcGIS Pro 3.4 to build the network cost matrix for accessibility calculations.

3.2.3. Equality analysis

For comparison, we also calculated conventional, capacity-based walking accessibility, treating the cumulative area as supply for the ESFCA. Spatial equality was then assessed using the Lorenz curve and its companion Gini coefficient, which are classic measures in green accessibility research (Liang et al., 2023). The Lorenz curve plots the cumulative share of walking accessibility reached by the poorest-served percentage of residents; its bow away from the 45° line (perfect equality in perceptions/qualities) of equality visualises inequality. The formula for the Gini Index is as below:

$$GiniIndex = 1 - \sum_{i=1}^n (P_i - P_{i-1})(A_i + A_{i-1}) \quad (5)$$

where P_i denotes the cumulative population proportion, A_i refers to the cumulative proportion of the walking accessibility score. Notably, a Gini index ranges from 0 to 1, with 1 implying perfect inequality.

3.2.4. Bivariate LISA for spatial insights

Beyond global metrics, we explored spatial clustering analyses to examine horizontal equality, with Moran's I statistics confirming spatial autocorrelation in perception-based accessibility. We then applied

bivariate LISA to identify neighbourhoods where park accessibility and population are coupled to locate areas that are systematically over- or underserved (Adorno et al., 2025; Liang et al., 2023). LISA was further conducted to correlate perception-based and capacity-based accessibility, revealing “quality-quantity” mismatches, an overlooked form of green inequality in existing planning. Unlike composite indices that blend size and sentiment (Zou et al., 2024), this two-layer approach keeps both dimensions separate, sidestepping weighting biases while delivering novel guidance for planners.

3.3. Exploring factors that influence park perceptions

3.3.1. Collecting independent variables

To uncover the determinants of park perceptions through statistical analysis for concrete design and management suggestions in high-density contexts, we compiled a comprehensive set of spatiotemporal variables drawn from multisource data such as official datasets and street views (Table 1). Notably, this selection was guided by a synthesis of prior empirical and review studies highlighting that both park-internal design features and surrounding environmental characteristics jointly shape users' park experiences (McCormack et al., 2010; Roberts et al., 2019; Stessens et al., 2020; Xing et al., 2020).

Park internal attributes capture the physical form, layout, and facilities that underpin initial design quality and directly influence park use and satisfaction (Kaczynski et al., 2008; Roberts et al., 2019). Following methodologies established in prior park studies (Fan et al., 2021; Kong et al., 2022; Zhan et al., 2024), we derived indicators such as park size, proportion of built-up area, internal road intersection density, water ratio, and the landscape shape index (LSI)—a metric of spatial complexity linking area and perimeter. Drawing from Google Place data, we further included overall rating and binary indicators for provision of picnic areas, food services, toilets, sports facilities, and dog-, child-, or wheelchair-friendly design that signal how well a space accommodates the needs of diverse user groups. These park amenities have been reported to significantly impact park satisfaction (Roberts et al., 2019) and can influence actual use (McCormack et al., 2010).

Regarding the surrounding macro-level built environment (BE) attributes, we delineated a 5-minute walking buffer (400 m) around each park (Liu et al., 2024) and calculated the density of bus stops, metro entrances, and road intersections to characterise connectivity (Fan et al., 2021), which can affect the daily use of parks (McCormack et al., 2010). To represent land-use diversity and neighbourhood vibrancy, we computed the density of surrounding POIs (Kong et al., 2022), represented by the density of accommodation, restaurants, cultural, commercial, and educational facilities, while land use mix was summarised using the entropy index (Appendix E).

Micro-level street environment (SE) characteristics further refine the analysis by addressing micro-scale experiential qualities that influence perceptions and duration of park use (de la Barrera et al., 2016; Moran et al., 2020; Zhou et al., 2022). Street-level conditions were captured by sampling Google Street Images every 50 m along pedestrian networks within the 400 m buffer. The Mask2Former model (Cheng et al., 2022), pre-trained on ADE20K images, was adopted for semantic segmentation to extract Green View Index and Sky View Factor. Perceptual qualities of these streetscapes—safe, beautiful, lively, wealthy, boring, and depressing were predicted with the Place Pulse 2.0 dataset using the ResNet-50 architecture (Dubey et al., 2016) and were operated using the ZenSVI package in Python (Ito et al., 2025), resulting in continuous scores ranging from 0 to 10.

Additionally, sociodemographic context was integrated via census data for educational attainment, proportion of seniors, share of locally born residents, and median monthly household income. These indicators are associated with park perceptions and preferences across socioeconomic groups (Adorno et al., 2025; Moran et al., 2020; Roberts et al., 2019; Schüle et al., 2017; Sia et al., 2023). Finally, daily weather conditions—mean temperature, rainfall, relative humidity, sunshine hours,

Table 1
Summary of variables.

Var. name	Description	Source	Mean	S.D.	Max	Min
Dependent variables						
accessibility	Perceived Accessibility	Predicted using LLM	1.756	0.502	2	0
usability	Usability		1.451	0.592	2	0
attractiveness	Attractiveness		0.944	0.467	2	0
Independent variables						
Park Intrinsic Attributes						
dog_friendly_dummy	Whether Dog-friendly	Google Place	0.323	0.469	1	0
facility_sports_dummy	Sport facilities		0.278	0.45	1	0
facility_washroom_dummy	Public washroom		0.848	0.36	1	0
facility_picnic_dummy	Picnic facilities		0.0949	0.294	1	0
facility_children_dummy	Whether Children-friendly		0.494	0.502	1	0
onsite_service_dummy	On-site service		0.152	0.36	1	0
facility_wheelchair_dummy	Wheelchair friendly facilities		0.949	0.22	1	0
review_rating	Park rating		4.02	0.26	5	3.3
Area_sqm	Park Size (m ²)	Land use map	43,292	56,004	549,296	1,873
Bld_sqm_r	Internal Building Ratio	iG1000 Map	0.0255	0.0284	0.165	0
Int_inters	Internal road intersection density (quantity/m ²)	3D network	0.0134	0.00983	0.0543	0
Water_r	Water Area Ratio to the Park Size	OSM	0.0151	0.0336	0.244	0
LSI	Landscape Shape Index		0.501	0.161	0.868	0.14
Surrounding Macro-level BE (aggregated in 400 m Buffer around park boundaries)						
bld_ar_ra	Building density (m ² /km ²)	iG1000 Map	153,099	68,249	327,173	14,107
rdint_ra	Road intersection density (quantity/km ²)	3D network	0.0505	0.0256	0.141	0.0068
bus_ra	Bus station density (quantity/km ²)	iGeoCom 4.0	7.02	5.31	29.5	0
rail_ra	Subway station density (quantity/km ²)		4.72	6.56	36	0
acc_ra	Accommodation density (quantity/km ²)		22.9	99.3	751	0
resta_ra	Restaurant density (quantity/km ²)		127	160	969	0
cul_ra	Cultural facility density (quantity/km ²)		5.8	5.32	38.8	0
comm_ra	Commercial facility density (quantity/km ²)		19.7	16.3	88.5	0
edu_ra	Educational facility density (quantity/km ²)		22.3	15.1	73.8	0
landuse_mi	Land Use Entropy		0.23	0.0447	0.32	0.0819
Surrounding Micro-level SE (aggregated in 400 m Buffer around park boundaries)						
GVI_mean	Green View Index	Google Street View Images	0.16	0.0777	0.436	0.0327
SVF_mean	Sky View Factor		0.115	0.0425	0.319	0.0195
safer_mean	Safer	Predicted based on the trained Place Pulse 2.0 dataset	5.36	0.214	5.88	4.75
more_beaut	More Beautiful		4.85	0.379	6.12	4.01
livelier_m	Livelier		5.32	0.44	6.34	3.93
wealthier_	Wealthier		5.75	0.303	6.49	4.75
more_borin	More Boring		4.71	0.349	5.64	3.87
more_depre	More Depressing		5.48	0.324	6.85	4.83
Sociodemographic						
eld_ratio_	senior (age > 65) population ratio	HK 2022 Census by Large Subunit Group (LSUG)	0.187	0.0357	0.254	0.0625
psec_ratio	Ratio of people who received post-secondary education	level	0.351	0.0929	0.567	0.152
local_rati	Ratio of people born in HK		0.61	0.0823	0.821	0.4
ma_hh_num_	Median monthly household Income		39,912	17,306	130,785	15,240
Weather						
temp_c	Temperature in Celsius	Hong Kong Observatory Data	24.2	5.01	32.4	4.9
rain_mm	Precipitation in millimetres		6.58	22.1	425	0
sun_hrs	Sunlight hours		5.168	3.85	12.2	0
humid_rela	Relative humidity		77.5	10.3	98	28

and seasonal dummies (spring, summer, autumn)—were incorporated to account for temporal variability in explaining park perceptions. Climatic and seasonal variations have been shown to influence both activity intensity and amenity preferences (Roemmich, 2014), offering an important temporal dimension for understanding perception dynamics.

3.3.2. Understanding the impact of environmental characteristics

To examine the relationship between environmental variables and perceptions, we conducted a series of statistical analyses. First, Variance Inflation Factors (VIFs) were calculated among independent variables, and we discarded variables showing high multicollinearity (> 10) to avoid spurious significance in regression. We then modelled each perception at the review level as positive (1) or negative (0) using generalised linear mixed models (GLMM), specifying park identity as a random intercept to reflect the nested structure of park comments. Lastly, perceptions were aggregated by day (excluding dates with fewer than 10 reviews) and regressed on temporal factors using Ordinary Least Squares (OLS) linear regression. The GLMM and OLS regression analyses comprehensively account for how various factors shape multi-

dimensional perceptions.

4. Results

4.1. Comparison of model performances

Table 2 presents the performances of sixteen ML models across three perceptions. Resembling prior studies (Zhao et al., 2024), the multi-modal encoder models demonstrate reliable performance (accuracy 73%–81%), yet are outperformed by text-encoders and LLMs. Notably, assigning more weight to text than review image results in slightly increased accuracy. However, the improvement of multimodal encoders over unimodal text encoder models is inconclusive. The CLIP-text encoder model outperforms the multimodal models on perceived accessibility and achieves comparable strengths on other perceptions, implying that the information gain from images may be ineffective, at least in our case.

Our custom-developed Park-Perception-LLM exhibits substantial improvement over multimodal encoder models, specifically CLIP-based

Table 2

Model accuracy.

Model	Perceived Accessibility	Usability	Attractiveness
Multimodal encoder models			
CLIP MHA 1:1	0.757	0.769	0.800
CLIP Concat 1:1	0.758	0.732	0.809
CLIP MHA 2:1	0.754	0.763	0.794
CLIP MHA 4:1	0.762	0.769	0.812
Unimodal text encoder models			
CLIP-text encoder	0.773	0.768	0.794
DeBERTa	0.798	0.779	0.879
OpenAI text-embedding v3-large	0.812	0.803	0.831
LLMs (Zero-shot prompt)			
GPT-4o mini	0.818	0.754	0.790
GPT-4o	0.816	0.754	0.788
DeepSeek v3	0.818	0.727	0.739
DeepSeek r1	0.811	0.752	0.790
Grok-2	0.821	0.765	0.772
LLMs (Few-shot prompt)			
GPT-4o mini	0.850	0.757	0.827
GPT-4o	0.832	0.754	0.844
DeepSeek v3	0.813	0.778	0.808
DeepSeek r1	0.804	0.742	0.821
Grok-2	0.824	0.785	0.819
LLMs (Fine-tuning)			
Park-Perception-LLM (Fine-tuned GPT-4o mini)	0.867***	0.834***	0.911***

Note: *** denotes the best-performed model.

models, achieving over 10% higher accuracy across all dimensions, highlighting its superiority in predicting park perceptions. Specifically, it attains accuracies of 86.7% for perceived accessibility, 83.4% for usability, and 91.1% for attractiveness. Furthermore, as presented in Table 3, the 5-fold cross-validation results— $86.3\% \pm 1.7\%$ for perceived accessibility, $85.1\% \pm 1.6\%$ for usability, and $88.8\% \pm 1.4\%$ for attractiveness (mean $\pm$ SD)—are consistent with the previously reported test set accuracy, indicating minimal performance variance across data partitions. This consistency supports that Park-Perception-LLM is robust and stable and exhibits strong potential for generalising on unseen data.

Additionally, for perceived accessibility, the few-shot GPT-4o mini and GPT-4o models achieve high accuracy of 85.0% and 83.2%, respectively, outperforming zero-shot and other LLMs. For usability, the OpenAI text-embedding model ranks second best (80.3%), closely followed by the few-shot Grok2 (78.5%). Regarding attractiveness, DeBERTa and the few-shot GPT-4o rank second (87.9%) and third (84.4%), respectively.

4.2. Analysis of park perceptions

A large dataset of park perceptions was obtained by applying the Park-Perception-LLM to reviews. Approximately 21% of the reviews ($n = 14,409$) discussed perceived accessibility, with 12,082 comments (83.9%) being positive, and 2,327 (16.1%) being negative. Comments regarding usability were more prominent, representing nearly half of the dataset (49.8%), with 30,558 positive and 3,551 negative

Table 3

Five-Fold Cross-Validation accuracy of the Park-Perception-LLM.

Fold	Accessibility	Usability	Attractiveness
Fold 1	0.870	0.831	0.909
Fold 2	0.852	0.868	0.893
Fold 3	0.855	0.839	0.878
Fold 4	0.890	0.862	0.875
Fold 5	0.849	0.859	0.886
Average (5-fold CV)	0.863	0.851	0.888

classifications. For attractiveness, a small fraction of reviews was deemed irrelevant (8.1%), while 78.0% expressed positive views and 13.9% were negative. These findings illustrate an overall favourable public perception of the usability and attractiveness of parks, highlighting the moderate yet significant concerns regarding accessibility in Hong Kong. The temporal patterns are detailed in Appendix F.

Fig. 4 illustrates the spatial distribution of perceptions at the park and district levels. Given Hong Kong's dense urban form, urban parks tend to cluster within core urban areas. Regarding perceived accessibility, most parks are rated high; however, lower values are primarily observed in Kowloon, Hong Kong Island, and to a lesser extent, in Tuen Mun in the western New Territories. Regarding usability, parks with lower ratings are predominantly concentrated in Kowloon City, Kwun Tong, and Wong Tai Sin. A similar pattern is evident for attractiveness, with lower-rated parks clustered in Kowloon and adjacent areas such as Kwai Tsing and Eastern Hong Kong Island.

4.3. Park accessibility and equality analysis

Figs. 5a and 5b visualise the perception-weighted park accessibility for 800 m and 1,200 m buffers. High accessibility is consistently concentrated in the densest urban cores—Central, Wan Chai, and the Eastern District on Hong Kong Island, as well as central Kowloon (Yau Tsim Mong and parts of Sham Shui Po). Accessibility falls sharply southwards on the island and towards Kwun Tong's eastern fringe and Sham Shui Po's western fringe. Notably, pockets of moderate-to-high accessibility appear in Tsuen Wan, Sha Tin, and, more sporadically, in Tuen Mun. Most peripheral areas, especially the northern and north-eastern New Territories, remain inaccessible to quality parks.

Expanding the catchment from 800 m to 1,200 m enlarges the spatial footprint of high scores by relaxing distance constraints. District-level gains are most pronounced in zones adjacent to high-access clusters. Overall spatial patterns, however, mirror those produced by conventional capacity-based E2SFCA scores (Figs. 5c & 5d). The few locations where perception-weighted and capacity-based results diverge highlight potential mismatches between the quantity of parks and their perceived quality, an issue explored further below.

The Lorenz curve and Gini-coefficient analysis (Fig. 6) confirm pronounced horizontal inequality. Under the 15-minute scenario, Gini values are 0.656 (perception-based) and 0.698 (capacity-based), while both exceed 0.75 in the 10-minute scenario. The top decile of residents enjoys more than half of the total park accessibility at 10 min and roughly 60% at 15 min. In contrast, the lower three-quarters of the population share less than 20% (800 m) or 30% (1,200 m) of the cumulative access. Thus, both size and perception-based accessibility metrics reveal gaps that persist across Hong Kong.

4.4. Mismatch between park accessibility and demand

Fig. 7 presents bivariate LISA maps juxtaposing perception-based park accessibility with population. The two distance thresholds yield broadly similar patterns, although the latter slightly enlarges the coverage of several clusters, signalling modest shifts in the spatial balance between supply and demand.

High-High clusters indicate areas with high-quality park accessibility and population density, representing ideal spatial coherence. They are limited in quantity and concentrated in the urbanised corridors, notably parts of Kwun Tong, Sham Shui Po, and Sha Tin, where park qualities match local demand. In contrast, Low-Low clusters also display coherence but imply an undesirable circumstance where both indicators are low. These swaths dominate peripheral and rural districts, particularly southern Hong Kong Island and vast stretches of the northern New Territories.

Meticulous attention should be paid to identifying critical spatial outliers for policy implications. Low-High clusters indicate problematic areas where dense populations face poor access to quality parks,

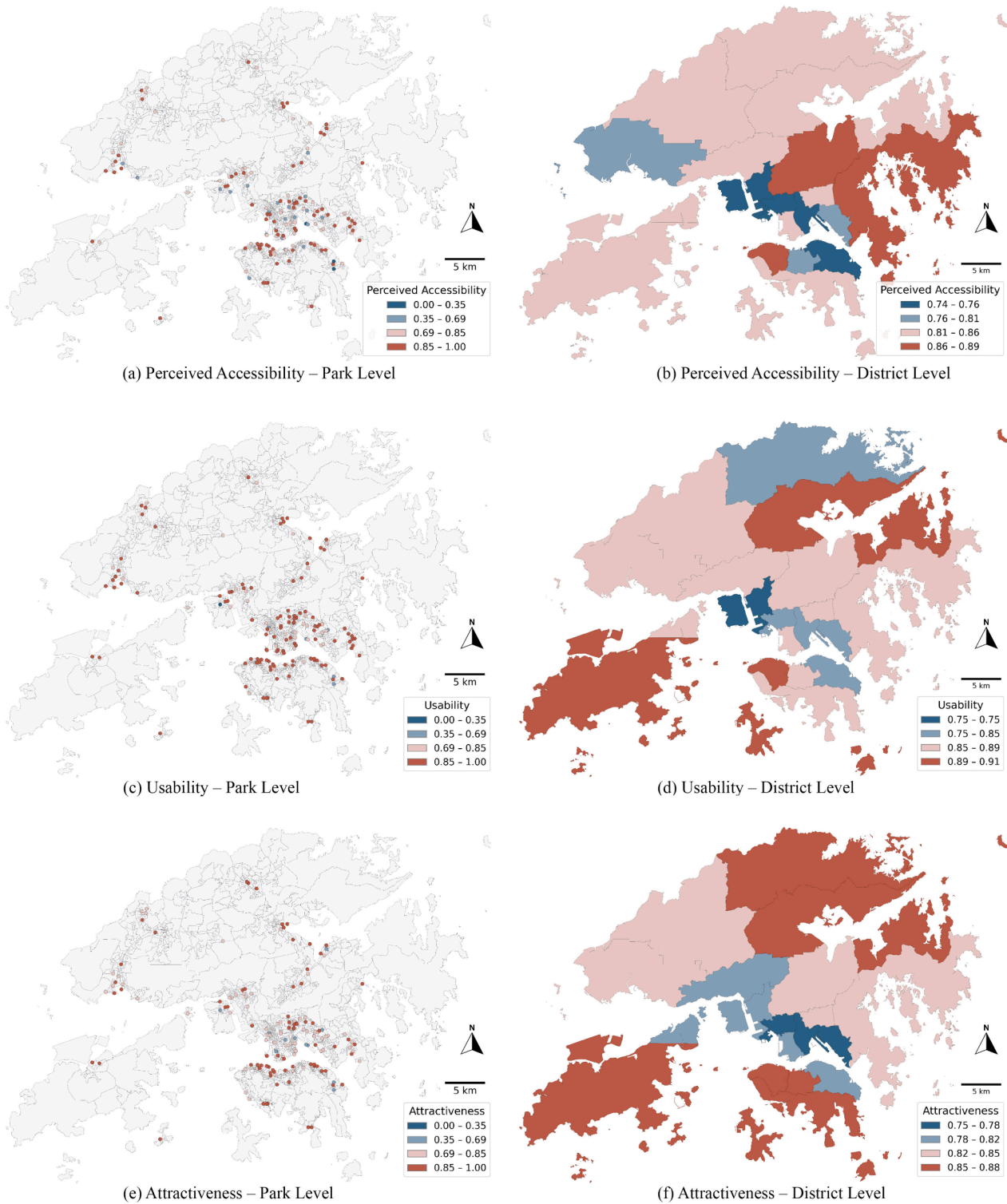


Fig. 4. Spatial distribution of park perceptions.

highlighting substantially unmet expectations. Pronounced L-H hotspots emerge across Tsuen Wan, Tuen Mun, Yuen Long, eastern Kwun Tong, much of Sha Tin, and the Southern District of Hong Kong Island. Conversely, High-Low clusters represent areas potentially leading to underutilisation, which are scattered in urban districts in Kowloon and the north of Hong Kong Island, suggesting less efficient allocations of park resources here.

When the walking threshold expands to 1,200 m, low-high (L-H) areas become more pronounced in Sha Tin, Tuen Mun, and Yuen Long.

This suggests that although residents can physically access more park area, their experiences remain poor due to limited facility diversity, low maintenance standards, and disconnection from residential clusters by major roads or terrain barriers. Conversely, high-low (H-L) clusters on southern Hong Kong Island and parts of Tuen Mun indicate well-maintained but spatially constrained parks, where good management offsets limited size. These areas risk future capacity pressure under densification, highlighting the need for integrated open space planning that balances quantity and quality.

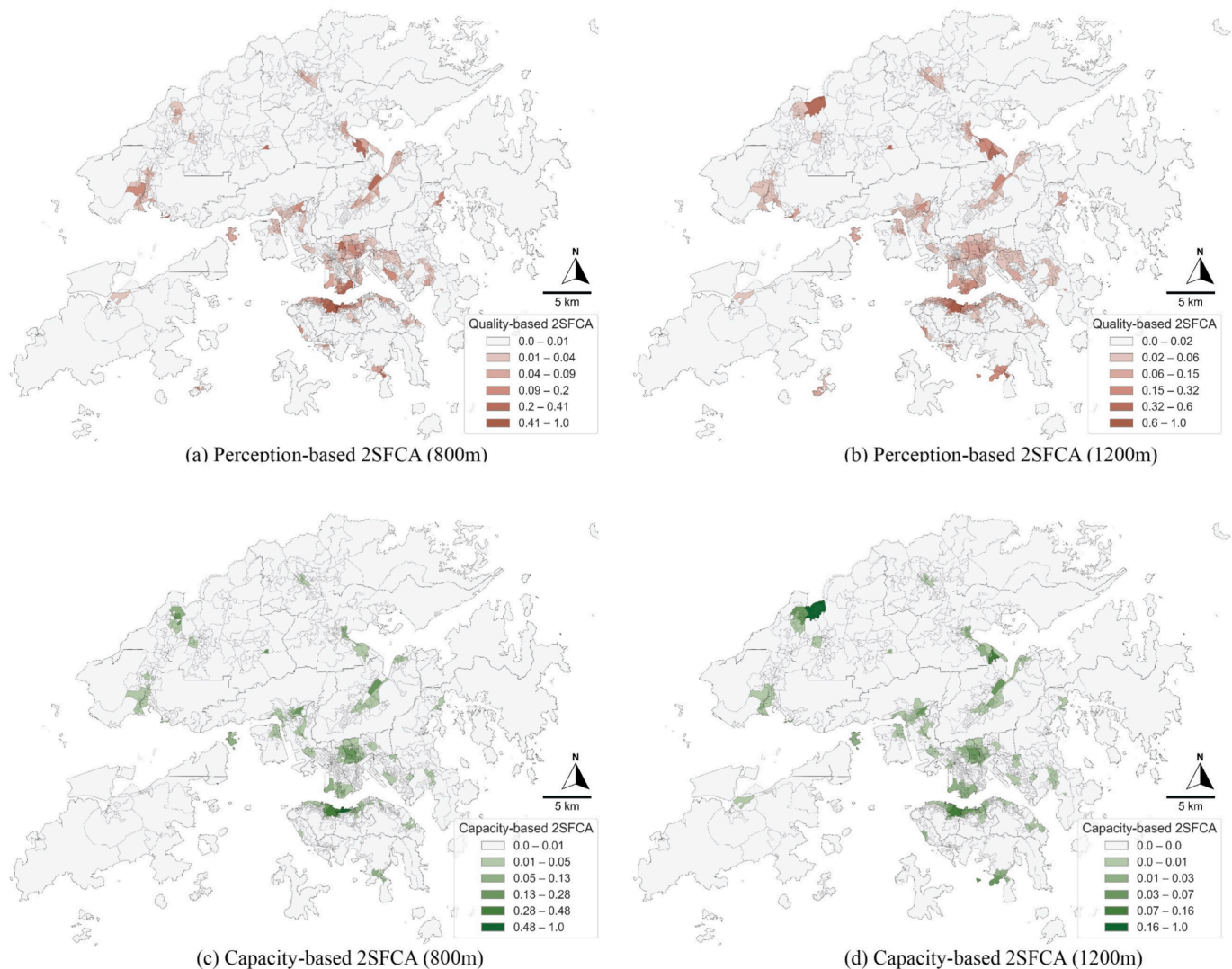


Fig. 5. E2SFCA park accessibility.

4.5. Mismatch between perception- and capacity-based park accessibility

Bivariate LISA allows us further to visualise spatial coherence and divergence between quality-based (perceptions) and capacity-based (size) park accessibility (Fig. 8) in the walking mode, which derives novel insights by exposing where the perceptions of parks align or conflict with their sheer quantities. Across both 800 m and 1,200 m walk buffers, pronounced disparities emerge. Besides the high perception–high capacity clusters present around central Kowloon and pockets in northern Hong Kong Island, low-low clusters dominate large swathes of the northern New Territories, parts of Tai Po Yuen Long, and Tseung Kwan O—signalling a persistent shortfall in both capacity and perceived standards in these new towns.

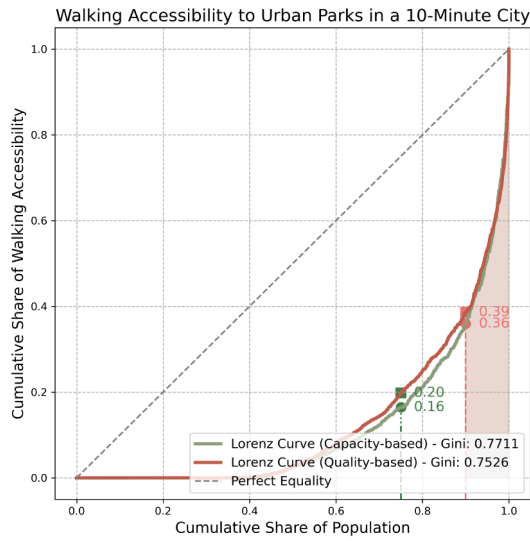
When the distance threshold expands to 1,200 m, low–high (L-H) clusters become more prominent in Sha Tin, Tuen Mun, and Yuen Long. This scale sensitivity arises because a larger catchment radius broadens the set of accessible parks, integrating adjacent neighbourhoods and altering local spatial associations between capacity and perceived quality. This pattern indicates that although residents can technically access a greater total park area, their experiences remain unsatisfactory. The underlying causes may include limited diversity of facilities, inadequate maintenance, and perceived accessibility costs arising from poor pedestrian connectivity, such as major road separations or terrain barriers. For example, residents of Mong Tseng Tsuen and Kak Tin Tsuen in

Yuen Long share physical access to the nearby Hong Kong Wetland Park, which is a large-scale green open space. Nevertheless, its relatively low perceived accessibility and moderate attractiveness score suggest the need for improvements to strengthen everyday usage and user satisfaction for green equality.

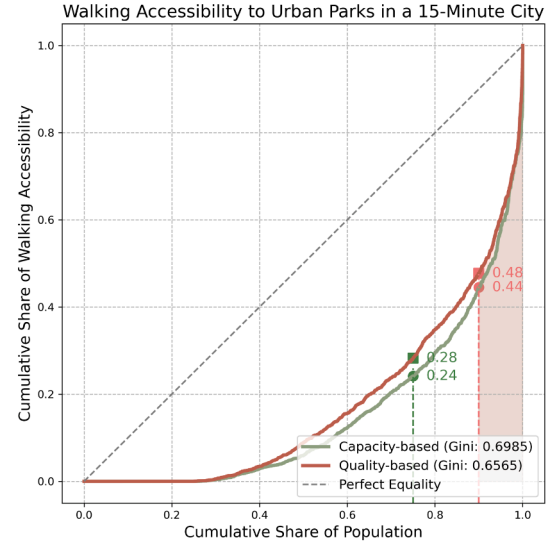
Conversely, High-Low clusters appear only sporadically (southern Hong Kong Island, north of Islands, and small areas of Tuen Mun and Sha Tin). These areas indicate residents can access well-maintained but spatially constrained parks, where good management offsets limited size. These areas risk future capacity pressure under ongoing urban densification and development, highlighting the need for integrated open space planning that balances quantity and quality. When the provision of large urban parks is infeasible, measures such as enriching street vegetation, creating pocket green spaces as minor open spaces, and establishing visually connected green corridors can extend the functional and perceptual benefits of parks into the dense urban fabric (Hursh et al., 2024). The cross-scale interpretation underscores that park accessibility inequalities are not static but depend on walking distance and residents' mobility ranges, and therefore should be jointly and comprehensively evaluated.

4.6. Impact of spatial factors

Moving beyond the interpretations of spatial patterns, we employed

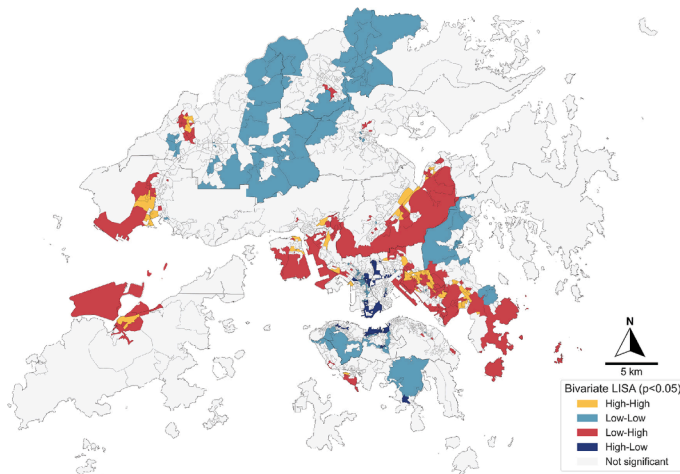


(a) 800m distance

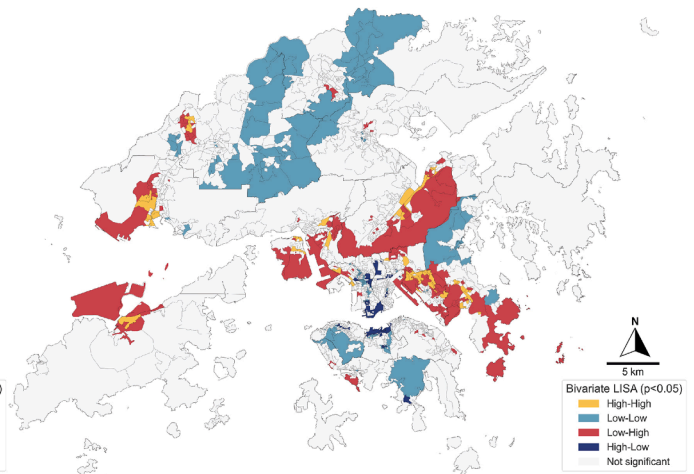


(b) 1,200m distance

Fig. 6. Lorenz Curve Plots.



(a) Accessibility & Population (800m)



(b) Accessibility & Population (1200m)

Fig. 7. Bivariate LISA maps of park accessibility and population demand.

GLMMs to elucidate the significant environmental and social determinants influencing park perceptions. Appendix G reports GLMMs with park intrinsic attributes only, while Table 4 presents the augmented models with surrounding built-environment and sociodemographic covariates. The composite models show modest gains in explanatory power (slightly higher marginal and conditional R^2) in all perception dimensions, and intrinsic park qualities account for most of the explained variance, with contextual surroundings offering Supplementary improvements.

Usability achieves the highest Conditional R^2 (0.369), followed closely by perceived accessibility (0.360) and attractiveness (0.341). The Marginal R^2 , representing the variance explained solely by fixed effects, is 0.351, 0.296, and 0.325, respectively. The difference between Conditional and Marginal R^2 reflects the contribution of random effects associated with park identity, with perceived accessibility demonstrating the most significant disparity (0.064) between Marginal and

Conditional R^2 . It suggests that it is significantly influenced by the inherent geographic characteristics, thereby highlighting considerable park heterogeneity.

The odds ratios of significant variables from the GLMMs are visualised in Appendix H. For perceived accessibility, park attributes such as children's facilities, onsite services, water ratio, and review ratings show significantly positive impacts. Additionally, surrounding environmental characteristics, including cultural facilities and greater street greenery, yield positive effects. Conversely, park size is negatively associated with perceived accessibility.

Regarding usability, sports facilities, washrooms, children's amenities, onsite services, and ratings exhibit positive influences. Enhanced street greenery and neighbourhoods with higher educational status also positively contribute to usability. In contrast, land use diversity, micro-level street aesthetics, and household income negatively affect usability.

Finally, for attractiveness, positive associations are detected with

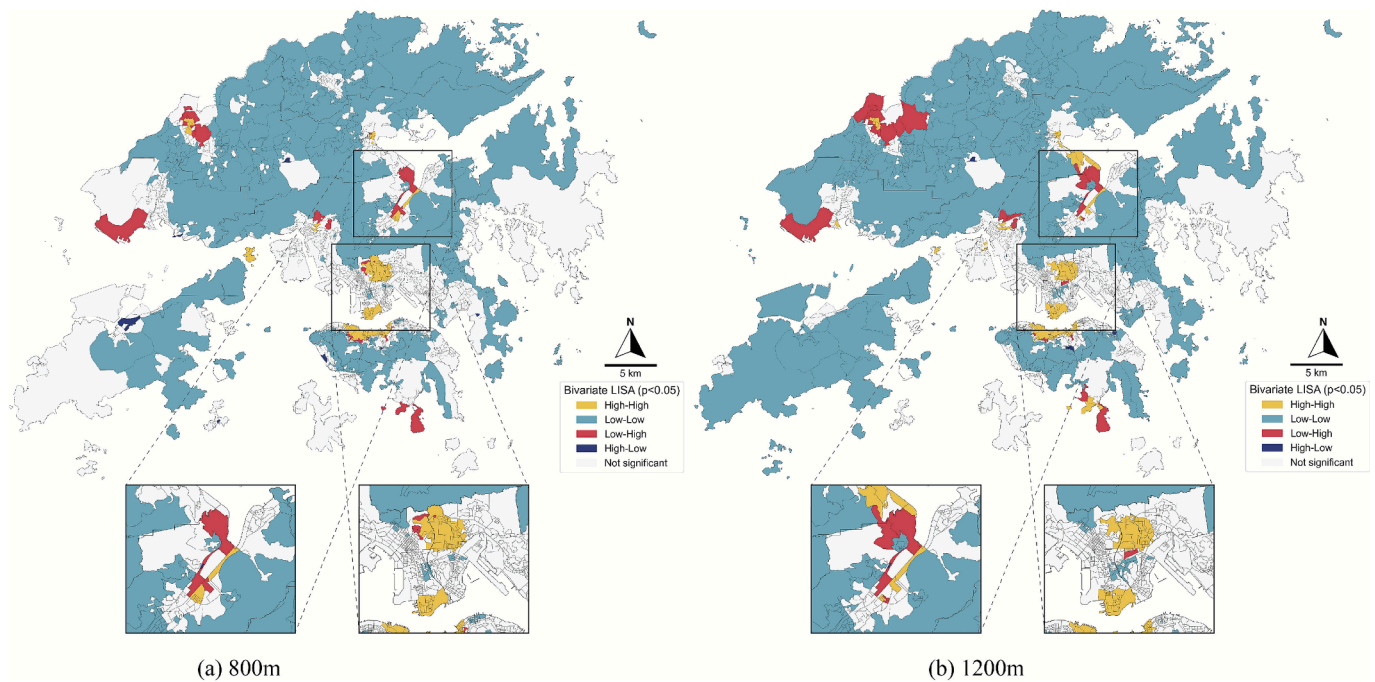


Fig. 8. Bivariate LISA clusters of Perception- & Capacity-based E2SFCA.

Table 4

Generalised Linear Mixed Model (GLMM) Results.

Variable	Perceived Accessibility			Usability			Attractiveness		
	Coef.	OR	Sig.	Coef.	OR	Sig.	Coef.	OR	Sig.
(Intercept)	1.47	4.335	***	2.32	10.209	***	1.61	4.985	***
dog_friendly_dummy	-0.121	0.886		-0.092	0.912		-0.0775	0.925	
facility_sports_dummy	0.194	1.214		0.166	1.181	*	0.0933	1.098	
facility_washroom_dummy	0.132	1.142		0.176	1.193		0.489	1.631	***
facility_children_dummy	0.601	1.824	***	0.411	1.509	***	0.0799	1.083	
onsite_service_dummy	0.498	1.645	**	0.215	1.24	**	0.177	1.194	**
Area_sqm	-0.535	0.586	***	-0.157	0.854	**	-0.210	0.811	***
Bld_sqm_r	0.0219	1.029		0.0344	1.035		0.0702	1.073	**
Water_r	0.416	1.515	***	0.0683	1.071		0.0986	1.104	*
LSI	-0.0901	0.913		-0.00298	0.997		-0.00036	1.000	
review_rating	1.16	3.194	***	1.33	3.789	***	1.24	3.451	***
rail_ra	0.0332	1.034		-0.0851	0.995		-0.0365	0.964	
cul_ra	0.164	1.164	**	0.089	1.093	*	0.0639	1.075	
landuse_mi	0.0625	1.065		-0.140	0.869	*	-0.181	0.835	***
GVI_mean	0.267	1.308	*	0.253	1.516	**	0.253	1.288	***
SVF_mean	0.0589	1.061		0.0203	1.003		0.0633	1.074	
safer_mean	-0.101	0.904		-0.00405	1.004		-0.0198	0.980	
more_beaut	-0.163	0.851		-0.0906	0.746	***	-0.096	0.908	
more_borin	-0.0756	0.927		-0.165	0.903		-0.151	0.860	***
eld_ratio	0.023	1.023		0.0456	1.053		0.0578	1.059	
psec_ratio	0.105	1.111		0.187	1.206	***	0.175	1.191	***
ma_hh_num	-0.148	0.862		-0.121	0.886	*	-0.0947	0.910	*
N(Observations)	14,409			34,109			62,896		
Marginal_R2	0.295918			0.351236			0.324698		
Conditional_R2	0.359931			0.368501			0.341375		
AIC	9815.4			16333.6			39928.2		

Notes: OR: Odds Ratio; AIC: Akaike Information Criteria. * $p < 0.1$. ** $p < 0.05$. *** $p < 0.01$. All continuous independent variables were z-standardised.

facilities, including washrooms, onsite services, internal building ratio, and neighbourhood characteristics, including proportions of senior residents and higher educational status. Conversely, larger parks, increased land-use diversity, streetscapes perceived as boring, and greater density of nearby higher-income households are negatively associated with attractiveness.

4.7. Impact of temporal factors

Additionally, OLS regression was used to test whether daily and

seasonal conditions temporally inform visitors' perceptions. As summarised in Table 5, temporal characteristics add no explanatory power to the model for perceived accessibility, while significantly influencing usability and attractiveness. Higher temperature is statistically associated with lower usability and a larger decline in attractiveness. Additionally, seasonality matters only for attractiveness, with ratings falling significantly during winter. Rainfall, sunshine duration, and humidity level show expected negative signs, but do not attain statistical significance.

Table 5
OLS Results on Temporal Park Perceptions.

Variable	Perceived Accessibility (ln)		Usability(ln)		Attractiveness (ln)	
	Coef.	Sig.	Coef.	Sig.	Coef.	Sig.
const	−0.0335		−0.0816	***	−0.0482	**
Temp_Cel	−0.0005		−0.0029	***	−0.0044	***
Rain_mm	−0.0009		−0.0001		0	
Sun_hrs	0.0004		0.001		0.0013	
Humid_rela	−0.002		0.0004		−0.0001	
spring	0.0715		0.0049		−0.005	
winter	−0.0042		−0.0083		−0.0308	***
summer	−0.0456		0.004		0.0052	
N(Observations)	159		1324		2376	
R ²	0.085		0.011		0.017	
F Statistics	2.013		2.138	***	5.969	***
AIC	−166.8		−2517		−4230	

Notes: ** p < 0.05. *** p < 0.01.

5. Discussion

5.1. The effectiveness of fine-tuned LLMs

LLMs demonstrate clear advantages over conventional NLP approaches (e.g., bag-of-words, n-grams) in terms of predictive performance, generalisability, and ease of deployment (S. Luo et al., 2025). Our study demonstrates that domain-specific fine-tuning can transform a general-purpose LLM into an accurate park perception classifier based on text, demonstrating the feasibility of leveraging LLM to solve perception measurement challenges in urban planning literature.

Across our experiments, text-only encoders and LLMs perform consistently better than multimodal models such as CLIP in our experiment, where there might be two potential reasons. First, photos accompanying online reviews rarely align semantically with the review text, introducing noise rather than complementary information—unlike the tightly paired image–text data used during CLIP pre-training. Second, CLIP truncates longer text with a 77-token context window, causing information loss in longer reviews. In our contrastive experiment, larger text encoders such as DeBERTa can enhance prediction accuracy without any image, indicating higher significance of a greater text window over adding images.

Echoing recent findings in Computational Social Science (Ziems et al., 2024) and urban planning research (X. Fu et al., 2024), zero-shot LLMs pre-trained on broad, out-of-domain corpora can already deliver satisfactory performance on park perceptions. Notably, augmenting prompts with as few as 20 labelled examples in a few-shot setting further improves accuracy at negligible cost. This offers a data-efficient alternative to the labour-intensive development of large, expert-curated datasets typically required by text-encoder-based models. Nonetheless, our fine-tuned Park-Perception-LLM achieves the highest accuracy, demonstrating the value of domain adaptation for planning research. Appendix I provides a preliminary analysis of model error patterns, identifying key categories of misclassification and offering insights into the linguistic characteristics that future LLM fine-tuning efforts may further exploit.

In general, by replacing sequential processing and handcrafted feature engineering with parallel, global context modelling, transformer-based architecture enables LLMs to capture long-range dependencies and complex semantic relationships more effectively. This technical advancement in model structure, combined with large-scale self-supervised pretraining on massive and heterogeneous text corpora (e.g., web data, books, and articles), allows LLMs to learn rich statistical representations of language, consistent with empirically observed scaling laws, and confer strong generalisability (Brown et al., 2020). As a result, LLMs move beyond surface-level pattern matching toward a more functional, task-oriented approximation of semantic understanding.

These capabilities further allow LLMs to effectively interpret subjective nuances such as sarcasm, humour, and figurative contents that exists in human language, providing stronger inferential capacity (C. Song et al., 2025). This capability is particularly important for landscape perception research, where user responses may describe true park experiences indirectly, through examples, metaphors, or context-dependent expressions. LLMs leverage deep, distributed semantic representations to model latent associations between descriptive physical cues and higher-level subjective psychological responses, enabling more robust inference of perception-related outcomes. In addition, LLMs benefit from their ease of access via online APIs and relatively low deployment costs, eliminating the need for extensive local infrastructure, model training, or specialised hardware.

In short, LLMs hold significant promise in AI4X (AI for Science) studies and enable transformative advances for urban studies and planning: with its interdisciplinary knowledge and elaborate reasoning abilities, it can assist in decision-making for planning through conceptualisation-generation-evaluation pipeline (Zheng et al., 2025), gauge public sentiment on proposed plan revisions (X. Fu et al., 2024), and streamline survey workflows to deepen insights into human–environment interactions (Jansen et al., 2023). Embedding expert domain knowledge and common-sense reasoning within retrieval-augmented LLMs could further distil complex urban dynamics into actionable intelligence for practitioners and the public. Moreover, considering the replicability, cost and inductive bias associated with closed-source LLMs, benchmarking open-source alternatives and building fully reproducible, interpretable pipelines is essential for transparent LLM-based urban planning research.

5.2. Understanding the impact of environmental variables on park preferences

5.2.1. Park internal attributes

Our analysis establishes a clear hierarchy of attributes shaping park perceptions in Hong Kong. The availability of diverse amenities, especially children’s play equipment and on-site services, can significantly increase attractiveness, confirming that targeted facilities directly satisfy users’ needs. Moreover, children’s amenities exert the most considerable positive effect on usability, in tandem with on-site services and sports areas, which is consistent with evidence that amenities underpin the value of cultural ecosystem services (Kong et al., 2023), and induce affective feelings and drive foot traffic (Huai et al., 2023; Roberts et al., 2019).

Our study confirms that natural landscape elements, most notably water features (Guo et al., 2024), can further enhance positive sentiment (Huai & Van de Voorde, 2022), bring happiness and restorative benefits (Tang et al., 2023; Wan et al., 2021), and improve perceived accessibility to raise awareness of the park (de la Barrera et al., 2016). Contrary to earlier work suggesting that park users tend to “remember the whole” rather than specific parts (Wan et al., 2021), we find that important elements, including water features, children-oriented play areas and onsite services, boost perceived accessibility, raising the awareness of the park qualities. A higher park internal building ratio may imply that more services raise attractiveness in the long term, underscoring the importance of well-maintained facilities.

Interestingly, size effects are nuanced, with size negatively correlated with attractiveness, usability, and perceived accessibility, aligning with evidence that size alone does not boost physical activity levels (Kaczynski et al., 2008) and suggesting that existing larger parks may be located further away from communities and that their qualities are below expectations. Nevertheless, large parks may deliver emotional benefits depending on the specific park categories (e.g., ecological, comprehensive, community parks), their ratio of impervious areas (Kong et al., 2022), and supplied facility types to attract senior residents in Hong Kong (Wang et al., 2022). This scale–convenience trade-off due to physical constraints highlights the need to carefully balance open

space ratios with neighbourhood density to achieve green space supply for future development.

5.2.2. Surrounding environment

Our findings also underscore the importance of the macro-level built environment and the micro-level street environment surrounding urban parks in fostering a healthy city. Land-use mix does not positively affect attractiveness, suggesting that highly heterogeneous surroundings may not necessarily increase the perceived attractiveness of surrounding parks (Zhao et al., 2025), though such places are typically more accessible and may increase foot traffic. In contrast, surrounding cultural facilities boost perceived accessibility and usability, which can lead to higher visitation rates, especially among older adults in Hong Kong (Lau et al., 2021; Wang et al., 2022).

Interestingly, the perceived street sky ratio and safety do not influence any park perception, diverging from studies in Latin America, where safety is decisive (de la Barrera et al., 2016; Moran et al., 2020). Hong Kong's low crime context may attenuate this effect. By contrast, street greenery consistently enhances all perceptions, most strongly on usability and most significantly on attractiveness, aligning with previous findings that micro-scale environments influence perceived park accessibility (Rigolon et al., 2018). Conversely, monotonous street-scapes reduce park attractiveness, highlighting the need for engaging, varied street design that supports park use and elevates public sentiment.

5.2.3. Sociodemographic and temporal factors

Education status positively influences perception of usability and attractiveness. Moreover, median household income is negatively linked to usability and attractiveness, while showing an insignificant relationship with perceived accessibility, implying that wealthier neighbourhoods, though closer to parks, judge programming, quality, cultural values, or maintenance more critically (Tang, 2017), highlighting inequities that go beyond simple park availability. Affluent communities may treat basic green-space provision as a baseline and impose higher standards for excellence and upkeep (de la Barrera et al., 2016).

Additionally, temporality also matters. Attractiveness dips in winter, possibly due to colder weather, and may be attributed to the decline in floral colour and vegetative diversity, reducing aesthetic appeal (Wan et al., 2021). High temperatures further suppress positive perceptions, highlighting the value of micro-climate improvement measures in parks.

These empirical patterns offer valuable insights and concrete planning guidance. Prominent street greenery, customised amenities for children and older adults, rigorous year-round maintenance may address the common concern that Hong Kong's green spaces feel "arguably the same" (Lau et al., 2021), and further supported by providing features with cultural values. Climate-adaptive strategies like shaded seating, water features, and heat-tolerant planting palettes can help avoid thermal discomfort. Collectively, these data-driven insights offer insights for parks that are accessible, authentically welcoming, satisfying, and inclusive to all.

5.3. Planning implications

Harnessing SMD with LLMs equips planners with a cost-effective, citizen-centred toolkit for park design and management. Because park users generate these bottom-up data and they remain publicly accessible, they are inherently transparent and auditable, lowering institutional barriers to adoption and reinforcing democratic accountability. The extracted multidimensional perceptions capture experiential subtleties that small-sample surveys often miss, offering great potential to facilitate citizen-centric planning when such information can be meaningfully interpreted by planning officials (Kong et al., 2022). Nevertheless, despite its excellent scalability, SMD-based approaches should not be viewed as a substitute for formal participatory toolkits, which provide deep engagement with key stakeholders to enable co-creation.

We further introduced a perception-based park accessibility index gauging park accessibility from perceptions rather than park size or count. Results show that equitable access to quality parks is as vital as spatial coverage of park capacity (Zhang et al., 2022). Incorporating subjective, experience-oriented quality provides a valuable alternative to purely capacity-based, objective indicators and emotion-based indices. Equality diagnostics reveal significant inequalities in perceived park quality, which would be undetected when only quantity is assessed. These results support an empirical foundation for regenerative planning that prioritises experiential equality instead of expanding park acreage.

From a practical perspective, bivariate LISA analysis pinpoints neighbourhoods exhibiting both limited park capacity and low perceived quality, where targeted investments—such as improving amenity qualities—could yield the highest social return. In the longer term, integrating *peri*-urban parks, pocket parks, and green streets into a continuous accessibility network could enhance overall urban liveability and resilience. However, to provide case-specific and actionable guidance for direct implementation, further calibration with ground-truth data and participatory validation would be required to support the planning exercise.

In short, integrating LLM-driven park perceptions with accessibility measures can develop a scalable decision-support system that can steer equitable development investment, set maintenance priorities, and transparently communicate performance gaps to promote a data-driven, citizen-informed model of urban governance relevant to many management sectors. Although we only demonstrated the feasibility in Hong Kong, the proposed framework can be adapted to diverse global contexts with attention to local cultural and spatial conditions. Cultural norms shape how residents perceive and use parks—collective leisure and intergenerational activities are more common in East Asian cities, whereas Western contexts often emphasise solitude and aesthetic appreciation (Low et al., 2009; Seaman et al., 2010). Differences in urban form also matter: in compact, high-density cities, micro-scale environments and street greenery strongly influence park perceptions, while in low-density settings, distance and connectivity may play a larger role (Kaczynski et al., 2008; Rigolon, 2016). Accounting for these contextual variations can enhance the robustness and transferability of the framework, supporting comparative studies across different urban environments.

5.4. Limitations and future research

This study contains several limitations that should be acknowledged. First, as a proof-of-concept study, the taxonomy of park perceptions was not exhaustive. Other important dimensions (e.g., thermal comfort, maintenance, crowding) could be further analysed for a more holistic understanding of park experience. Second, this study explicitly focuses on urban parks for leisure purposes, excluding minor open spaces (e.g., sitting-out areas) and country parks in Hong Kong due to resource limitations. These categories warrant future investigation, although feasibility depends on sufficient social media data to ensure adequate review volumes and minimise bias from data sparsity. Third, our district-mean imputation strategy for parks containing insufficient comments may introduce bias because parks within the same district might differ substantially. When fine-grained park attributes (e.g., amenity counts, size, functional types) become available from other data sources, a similarity-based imputation using parks with comparable characteristics within the district would likely yield more accurate estimates. Fourth, our three-way classification (irrelevant/positive/negative) oversimplifies perceptions that likely vary by their intensity, which may lie along a spectrum from very negative to very positive. Future work could explore methods to capture this continuous gradation for more nuanced inference, similar to the development of the sentiment score. Fifth, due to resource constraints, our research relied solely on SMD, and we found that certain perceptions (e.g., perceived accessibility) may be relatively underrepresented in review text due to platform-specific expression

biases and topic salience, and future studies could incorporate complementary multi-source mobility data that can capture more comprehensive reflections on their park experience. Sixth, leveraging before–after designs would enable future studies to investigate how site-specific interventions influence the evolution of public perceptions in response to policy initiatives, construction projects, or renovations. Finally, due to its sampling bias, insights derived from social media should be interpreted cautiously regarding demographic and behavioural representativeness and can be further validated through interviews or surveys with randomly selected park users. To gain a more comprehensive understanding, park managers could also survey individuals who choose not to visit parks, thereby identifying potential barriers and pathways to promote park usage (Sia et al., 2023).

6. Conclusion

Leveraging user-generated social media data, this research introduces a comprehensive analytical framework that benchmarks state-of-the-art LLMs for public perception evaluation of urban parks. The study demonstrates that the domain-specific Park-Perception-LLM can achieve optimal performance, yielding accuracy between 83% and 91% across various perception categories. Moreover, zero-shot and few-shot prompted LLMs achieve reasonable accuracy, proving the feasibility of LLMs' application in urban planning and green space research. Quantitative and spatial equality analysis of perception-based metrics further sheds light on the quality perspective largely overlooked in prevailing capacity-driven park accessibility frameworks. Using bivariate LISA, spatial clusters of high-capacity yet low-perceived park accessibility were identified in several neighbourhoods in Hong Kong, a high-density city where sparse resources impact vulnerable groups significantly. This showcases the method's ability to identify improvement priorities based on quality-driven, data-informed analysis, pinpointing areas requiring prioritised resource allocation for equality-focused park development. Finally, the explanatory results offer evidence-based recommendations by highlighting significant park attributes and macro- and micro-level environmental characteristics associated with perceptions. For example, diverse and well-maintained internal facilities, natural park elements, situating nearby cultural amenities, and lush street greenery in the surrounding environment significantly enhance people's perceptions of park quality, whereas monotonous streetscapes can negatively affect a park's overall attractiveness. In summary, the proposed LLM-based analytical framework enables reliable inference of experiential qualities from online reviews, thereby advancing a transparent approach using data from citizen-centred platforms. By translating subjective perceptions into interpretable spatial insights, the framework contributes to the existing practice of social infrastructure and provides empirically grounded guidance for long-term open space management to enhance the quality of life.

CRedit authorship contribution statement

Qiwei Song: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing, Funding acquisition. **Siyu Tian:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation. **Lingwei Zheng:** Writing – review & editing, Visualization, Investigation, Data curation. **Yuxuan Zheng:** Visualization, Investigation, Data curation. **Lin Qiu:** Writing – review & editing, Supervision, Project administration. **Bo Huang:** Writing – review & editing, Supervision. **Jeroen van Ameijde:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the CUHK Computational Social Science Fellowship, CUHK Faculty Research Data Management Incentivising Programme Fund and CUHK Interdisciplinary Research Seed Funding 2025-26.

We gratefully acknowledge Tony Tam and Hai Liang of the CUHK Computational Social Science Laboratory (CSSL), as well as other faculty members and CSS fellows, for their invaluable guidance and support throughout this study. We also thank Yasmine Yue Wu for her kind assistance with the preliminary data collection.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.landurbplan.2025.105571>.

Data availability

Data will be made available on request.

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